Huawei Access Node KOR-ACC-83 Routing Protocol Flap Major Alarm (ALARM_7277) Standard Operating Procedure

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7277

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Huawei Access Node KOR-ACC-83 at Koramangala Datacenter is a complex failure domain that occurs due to the intricacies of routing protocol interactions and the underlying hardware mechanics. The Huawei Access Node, being a critical component of the network infrastructure, plays a pivotal role in maintaining the stability and integrity of the routing protocol ecosystem. The occurrence of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on this node can be attributed to various factors, including misconfiguration, hardware failures, or software glitches. The BGP route churn, in particular, can be caused by a cascade of events, starting from a faulty router advertisement or an incorrect routing policy, which can eventually lead to a route flap, causing instability in the network. Similarly, OSPF adjacency loss can occur due to a mismatch in OSPF parameters or a lack of adjacency establishment, while ISIS LSP flooding can be caused by a failure in the ISIS LSP database or an excessive number of LSPs being flooded. Understanding the root causes of these issues is crucial in resolving the alarm and restoring normal network operations.

Pre-requisite Checks

- Ensure remote hands are dispatched to Koramangala Datacenter for physical access to the Huawei Access Node KOR-ACC-83.

- Verify the redundant link status and ensure that the primary and secondary links are functioning correctly.

- Check the node's power supply and ensure that it is receiving stable power.

- Verify the node's temperature and ensure that it is within the recommended operating range.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policy	High
Faulty router advertisement	Medium
Hardware failure in the Access Node	Low
Software glitch in the routing protocol software	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp routing-table all, Expected Output: Display the BGP routing table with all routes, Description: This command displays the BGP routing table with all routes, including the local and remote AS numbers, the route origin, and the route path. This can help identify any misconfigured BGP routing policies or faulty router advertisements.

Command: display ospf neighbor, Expected Output: Display the OSPF neighbor table, Description: This command displays the OSPF neighbor table, including the neighbor's IP address, the state of the adjacency, and the number of LSA packets sent and received. This can help identify any OSPF adjacency loss or mismatch in OSPF parameters.

Command: display isis lsp, Expected Output: Display the ISIS LSP database, Description: This command displays the ISIS LSP database, including the LSP ID, the sequence number, and the LSP payload. This can help identify any ISIS LSP flooding or failure in the ISIS LSP database.
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Resolution Steps

Step 1: Step: Reset the BGP process, Command: reset bgp all, Description: Reset the BGP process on the Access Node to clear any misconfigured BGP routing policies or faulty router advertisements.

Step 2: Step: Reconfigure OSPF parameters, Command: ospf area 0.0.0.0 default-route 0.0.0.0 0.0.0.0, Description: Reconfigure the OSPF parameters on the Access Node to resolve any OSPF adjacency loss or mismatch in OSPF parameters.

Step 3: Step: Clear ISIS LSP database, Command: clear isis lsp all, Description: Clear the ISIS LSP database on the Access Node to resolve any ISIS LSP flooding or failure in the ISIS LSP database.

Post-Check Verification

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Command: display bgp routing-table all, Expected Output: Display the BGP routing table with all routes, Description: Verify that the BGP routing table is stable and that there are no misconfigured BGP routing policies or faulty router advertisements.

Command: display ospf neighbor, Expected Output: Display the OSPF neighbor table, Description: Verify that the OSPF neighbor table is stable and that there are no OSPF adjacency loss or mismatch in OSPF parameters.

Command: display isis lsp, Expected Output: Display the ISIS LSP database, Description: Verify that the ISIS LSP database is stable and that there are no ISIS LSP flooding or failure in the ISIS LSP database.
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Roll-back Procedures

- Step: Restore the BGP process, Command: restore bgp all, Description: Restore the BGP process on the Access Node to its previous state in case the resolution steps fail.

- Step: Reapply OSPF parameters, Command: undo ospf area 0.0.0.0 default-route 0.0.0.0 0.0.0.0, Description: Reapply the previous OSPF parameters on the Access Node in case the resolution steps fail.

- Step: Rebuild ISIS LSP database, Command: build isis lsp all, Description: Rebuild the ISIS LSP database on the Access Node in case the resolution steps fail.